

Homework 1 Solutions: Set Theory and Logic

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Practice Problems: Work together in class

1. Write each of the following sets in set-builder notation.

- (a) The set of all possible quantities q a firm can produce if output must be nonnegative is

$$\{q \in \mathbb{R} : q \geq 0\}.$$

- (b) The set of all prices p strictly greater than 0 and less than or equal to 100 is

$$\{p \in \mathbb{R} : 0 < p \leq 100\}.$$

- (c) The set of all income levels m that are at least 50 is

$$\{m \in \mathbb{R} : m \geq 50\}.$$

- (d) The set of all integer quantities q between 0 and 10, including both 0 and 10, is

$$\{q \in \mathbb{Z} : 0 \leq q \leq 10\}.$$

2. Let

$$A = \{1, 2, 3, 4\}, \quad B = \{3, 4, 5, 6\}, \quad C = \{2, 4\}.$$

- (a) The union contains all elements that are in A or in B :

$$A \cup B = \{1, 2, 3, 4, 5, 6\}.$$

- (b) The intersection contains all elements that are in both A and B :

$$A \cap B = \{3, 4\}.$$

- (c) Yes,

$$C \subset A.$$

This is because both elements of C , namely 2 and 4, are also elements of A .

- (d) No,

$$C \not\subset B.$$

This is because $2 \in C$, but $2 \notin B$.

- (e) Yes,

$$\emptyset \subset A.$$

The empty set is a subset of every set.

3. Suppose a consumer can choose bundles (x_1, x_2) of two goods, where both goods must be consumed in nonnegative amounts.

- (a) The consumption set is

$$X = \mathbb{R}_+^2 = \{(x_1, x_2) \in \mathbb{R}^2 : x_1 \geq 0, x_2 \geq 0\}.$$

(b) Yes,

$$(2, 5) \in X,$$

because both components are nonnegative.

(c) No,

$$(-1, 5) \notin X,$$

because the first component is negative.

(d) Negative consumption may be ruled out because, in many economic models, a consumer cannot consume a negative amount of a good. For example, one cannot eat negative apples or use negative units of housing. Negative quantities may represent selling or supplying, but that is a different economic activity from consumption.

4. Let

$$S = \{x \in \mathbb{R} : 10 \leq x \leq 50\}.$$

Interpret S as the set of feasible weekly labor hours for a worker.

(a) An upper bound for S is 50. Any number greater than 50 is also an upper bound.

(b) A lower bound for S is 10. Any number less than 10 is also a lower bound.

(c) The least upper bound of S is

$$\sup S = 50.$$

(d) The greatest lower bound of S is

$$\inf S = 10.$$

5. For each statement, write the converse and the contrapositive.

(a) Statement:

If a firm earns positive profit, then its revenue is greater than its cost.

Converse:

If a firm's revenue is greater than its cost, then the firm earns positive profit.

Contrapositive:

If a firm's revenue is not greater than its cost, then the firm does not earn positive profit.

(b) Statement:

If $x > 10$, then $x > 0$.

Converse:

If $x > 0$, then $x > 10$.

Contrapositive:

If $x \leq 0$, then $x \leq 10$.

(c) Statement:

If a consumer buys good 1, then the price of good 1 is affordable.

Converse:

If the price of good 1 is affordable, then the consumer buys good 1.

Contrapositive:

If the price of good 1 is not affordable, then the consumer does not buy good 1.

Homework Problems: Submit individually

6. Let

$$D = \{q \in \mathbb{R} : q \geq 0\}$$

and

$$E = \{q \in \mathbb{R} : q \leq 100\}.$$

(a) The intersection is

$$D \cap E = \{q \in \mathbb{R} : 0 \leq q \leq 100\}.$$

Economically, this is the set of output levels that are both nonnegative and do not exceed the firm's capacity.

(b) The union is

$$D \cup E = \mathbb{R}.$$

To see this, note that every real number either satisfies $q \geq 0$ or $q \leq 100$; indeed, these two sets overlap heavily.

(c) If the universal set is $\mathbb{R}$, then

$$D^c = \{q \in \mathbb{R} : q < 0\}.$$

(d) Yes, $D \cap E$ is bounded above. Its least upper bound is

$$\sup(D \cap E) = 100.$$

7. Consider the following sets:

$$A = \{x \in \mathbb{R} : x > 5\}, \quad B = \{x \in \mathbb{R} : x \geq 5\}, \quad C = \{x \in \mathbb{R} : x < 20\}.$$

(a) The intersection is

$$A \cap C = \{x \in \mathbb{R} : 5 < x < 20\}.$$

(b) The intersection is

$$B \cap C = \{x \in \mathbb{R} : 5 \leq x < 20\}.$$

(c) No, these are not the same set. The set $B \cap C$ contains 5, while $A \cap C$ does not.

(d) "Strictly more than 5 units" means quantities greater than 5, so 5 itself is not allowed. "At least 5 units" means quantities greater than or equal to 5, so 5 itself is allowed.

8. Let P be the statement "a consumer's income increases" and let Q be the statement "the consumer's feasible consumption set becomes larger."

(a) The statement $P \Rightarrow Q$ is:

If a consumer's income increases, then the consumer's feasible consumption set becomes larger.

(b) The converse is:

If the consumer's feasible consumption set becomes larger, then the consumer's income increases.

(c) The contrapositive is:

If the consumer's feasible consumption set does not become larger, then the consumer's income does not increase.

(d) The contrapositive is logically equivalent to the original statement. The converse is not logically equivalent to the original statement.

9. Determine whether each statement is true or false. Briefly explain.

- (a) True. If $A \subset B$, then every element of A is already in B . Therefore,

$$A \cup B = B.$$

- (b) True. If $A \subset B$, then the elements common to both A and B are exactly the elements of A . Therefore,

$$A \cap B = A.$$

- (c) False. If

$$x \in A \cup B,$$

then x is in A or in B , possibly both. It does not have to be in both sets.

- (d) True. If

$$x \in A \cap B,$$

then by definition $x \in A$ and $x \in B$.

- (e) True. The empty set is a subset of every set. Therefore, if $A = \emptyset$, then

$$A \subset B$$

for any set B .

- (f) False. The subset order is not always a complete order. Two sets may be incomparable. For example, if

$$A = \{1\} \quad \text{and} \quad B = \{2\},$$

then

$$A \not\subset B \quad \text{and} \quad B \not\subset A.$$

10. Suppose an economist claims:

If the price of a normal good falls, then the quantity demanded of that good rises.

- (a) Let

P : the price of a normal good falls

and

Q : the quantity demanded of that good rises.

- (b) The contrapositive is:

If the quantity demanded of that good does not rise, then the price of the normal good did not fall.

- (c) The converse is:

If the quantity demanded of that good rises, then the price of the normal good fell.

- (d) The converse is not automatically true because the converse is not logically equivalent to the original statement. The original statement says that a price fall is sufficient for quantity demanded to rise. It does not say that a price fall is the only possible reason quantity demanded can rise.
- (e) Quantity demanded might rise even if the price of the good did not fall because income increased, preferences shifted in favor of the good, the price of a substitute good increased, or expectations about future prices changed.